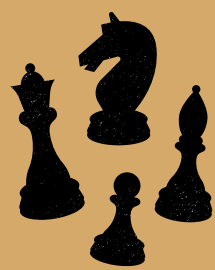


# Explaining Chess Positions Using Knowledge Representation



## Techniques

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## Motivation

- Chess engines: superhuman performance, limited explainability
- LLMs: natural explanations, but ungrounded claims.
- KR: strong relational structure, limited explainability



Chess Engine

Best Move: Nf3  
Evaluation: +1.43

Language Model

Nf3 is the best move because horses like fields



## Solution

- Represent chess positions using KR techniques.
- Interactive interface connecting a user to a knowledge base using a language model.



### Layer 1: Preprocessing

- FEN → coordinate matrices
- Datalog rules derive spatial chess features such as piece movement, open files/ranks, pawn structures, active pieces
- Facts → triples → ABox

```
% King Moves (Cannot walk into check)
legal_move("K", ?r1, ?f1, ?r2, ?f2) :-
    pseudo_legal("K", ?r1, ?f1, ?r2, ?f2),
    ~threatened_by_black(?r2, ?f2) .
```

```
closed_position(?id) :-
    pos_id(?id),
    blocked_center_file(4),
    blocked_center_file(5) .

open_position(?id) :- pos_id(?id), ~closed_position(?id) .
```

### Layer 2: Reasoning

- TBox axioms about chess concepts are merged with the ABox.
- A reasoner infers additional position-specific facts.

Description: WhiteControlledDiagonal

Equivalent To  
OpenDiagonal  
and (hasSquare some (Square  
and (hasPiece some (WhitePiece  
and (Bishop or Queen))))

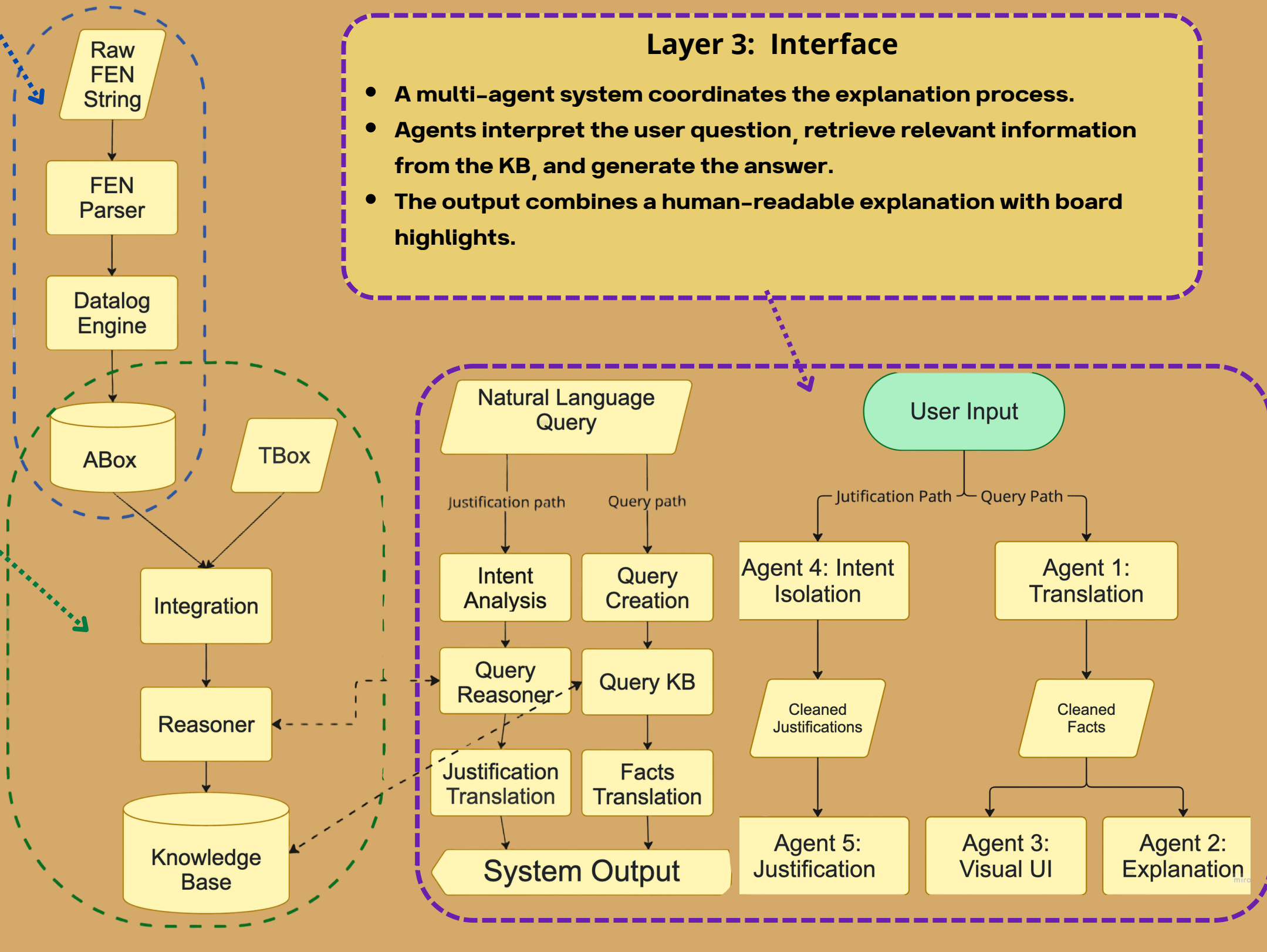
WhiteDisadvantage  
BlackInfiltrator  
WhiteBackwardPawn  
WhiteDoubledPawn  
WhiteExposedKing  
WhiteHangingPiece  
WhitePinnedPiece  
WhiteAbsolutePinnedPiece  
WhiteUncastledKing

Piece  
ActivePiece  
Bishop  
BlackPiece  
CenterControllingPiece  
HangingPiece  
King  
Knight  
OverWorkedPiece  
ParalysedPiece  
Pawn  
PinnedPiece  
Queen  
Rook  
WhitePiece

## Methodology

### Layer 3: Interface

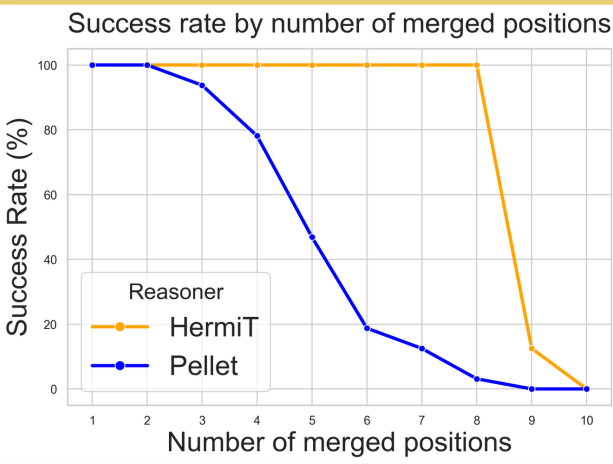
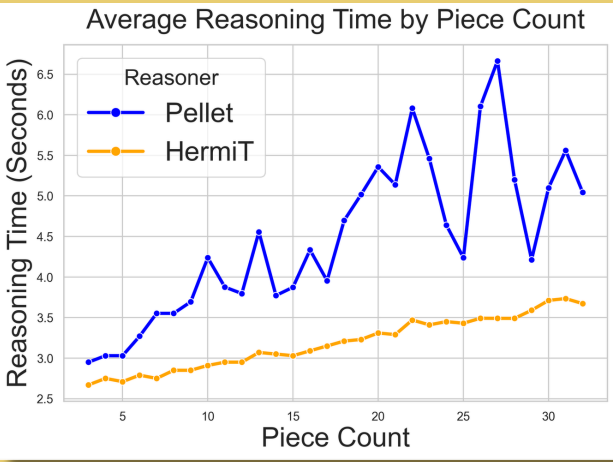
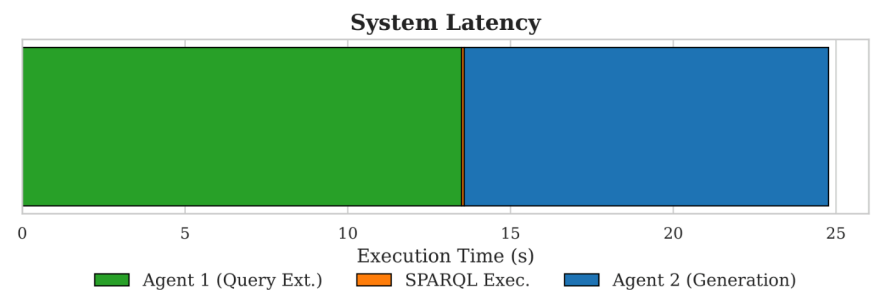
- A multi-agent system coordinates the explanation process.
- Agents interpret the user question, retrieve relevant information from the KB, and generate the answer.
- The output combines a human-readable explanation with board highlights.



## Evaluation

### System Performance & Scalability

- Longest latency comes from query extraction and answer generation.
- SPARQL execution is negligible compared to agent processing.
- HermiT outperforms Pellet in performance and scalability benchmarks.
- Piece count is positively correlated with reasoning time.



### User Study: Perceived Quality and System Preference

- Ontology-grounded system preferred for usefulness, factual accuracy, and confidence.
- Strongest effect: 9/9 participants judged it more factually accurate.
- Pure LLM was slightly preferred for naturalness.
- Likert scores were similar, suggesting comparable perceived quality.
- Main trade-off: grounded accuracy vs. natural language fluency.

| Measure           | Ontology-grounded | Pure LLM |
|-------------------|-------------------|----------|
| Understandability | 3.42              | 3.50     |
| Usefulness        | 3.33              | 3.19     |
| Trust             | 2.94              | 3.08     |
| Overall           | 3.23              | 3.26     |

| Question                | Ontology-grounded | Pure LLM |
|-------------------------|-------------------|----------|
| Most useful             | 5/8               | 3/8      |
| Most natural            | 4/9               | 5/9      |
| Most factually accurate | 9/9               | 0/9      |
| Most confidence         | 7/9               | 2/9      |

